EAOHP 2026 WORKSHOP MASTERCLASS

# *Using Actor-Partner Interdependence Models (APIM) to explore indistinguishable and distinguishable dyads with Multilevel (MLM) & Structural Equation Modeling (SEM) approaches.*

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*June 18, 2026 — University of Helsinki*

# Workshop Structure Overview

### 01. Foundations

*Understand dyadic dependencies, assumptions, key APIM parameters, and structural distinguishability.*

### 02. Indistinguishable

*Model exchangeable partners using Multilevel Modeling (lme4) and SEM equality constraints.*

### 03. Distinguishable

*Evaluate systemic roles (e.g. gender, hierarchy) using moderator, two-intercept, and multi-group SEM variations.*

### 04. Advanced Topics

*Address moderated crossover, diary designs, sample size guidelines, and missing data using FIML.*

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# What Is Dyadic Data?

*Dyadic data arise when observations are gathered in pairs (dyads) that demonstrate systematic interdependence.*

*Common examples include:*

- Romantic Couples: *Partners in heterosexual or same-sex relationships (often distinguishable).*
- Workplace Dyads: *Supervisor-subordinate, mentor-mentee configurations.*
- Parent-Child Pairs: *Mothers & daughters, fathers & sons.*
- Peer Groups: *Roommates, sports teammates, co-workers.*

*Non-independence violates classic OLS regression assumptions, leading to biased standard errors and highly inflated Type I error rates.*

## Simulated Workshop Dataset

*We utilize a double-earner dataset styled after* Hahn, Binnewies, & Dormann (2014, JVB)*:*

- Sample Size: *N = 100 couples (200 unique individuals)*
- Focal Predictor (X): *Work-Nonwork Connection (WNC)*
- Secondary Covariate (Z): *Psychological Recovery*
- Target Outcome (Y): *Relationship Satisfaction*
- Nesting Indices: *Gender, children indicators*

# Workshop Exercises

*An independent simulated dataset for practising crossover-of-work-outcomes techniques on a coherent research question.*

## Research Scenario

***How does a partner's affect, self-determination, and job crafting crossover to shape an actor's work outcomes?***

- ✓ **Predictors: *actor & partner affect, SDT, job crafting***
- ✓ **Outcomes: *engagement, performance, creativity***
- ✓ **Moderators: *live together, years together, time spent this morning***

## Dataset Specification

*N = 250 dual-earner dyads (500 individuals)*

- *Gender: male / female*
- *ICC(1) predictors ≈ 0.30*

*Files: exercise_data.Rdata ddl2 (long), ddw2 (wide)*

### Focal crossovers:

***partner job crafting x live_together -> engagement***

***partner SDT x years_together -> performance***

***partner affect x time spent -> creativity***

# Exercises & Answer Template

*A guided, prose-only workflow for adapting scripts 02-08 to the exercise dataset - emphasis on interpretation, not copy-paste.*

## exercises/exercises.md

*9 sections mirroring scripts 02-08 + integrative Section 9.*

- Sections 1-3: *scripts 02-04 (APIM, multilevel, distinguishable)*
- Sections 4-6: *scripts 05-07 (distinguishable SEM, two-intercept)*
- Section 9: *integrative reflection across all models on the exercise dataset*

## exercises/answers_template.R

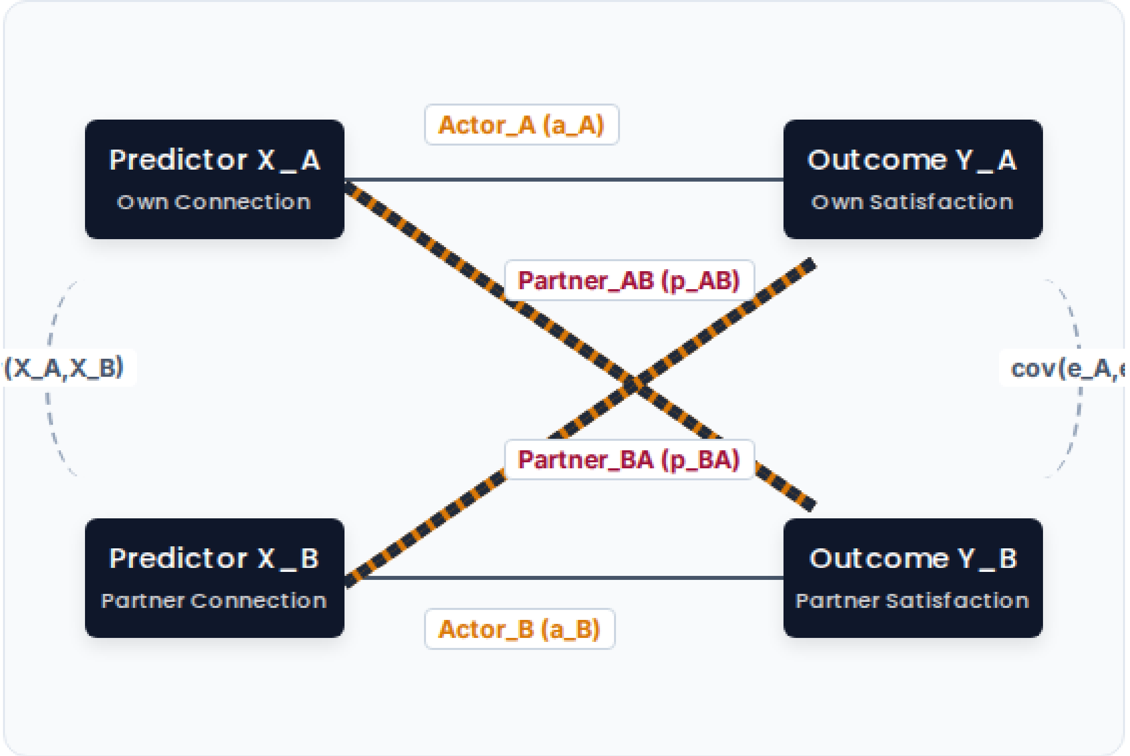
### How to use the template:

1. *Run simulate_exercise_data.R to load data*
2. *Read exercises.md section by section*
3. *Open answers_template.R alongside each script*
4. *Apply the substitution table to map data to scripts*
5. *Write a 1-sentence interpretation as a comment*

# The APIM Framework

*The **Actor-Partner Interdependence Model** (Kenny, Kashy, & Cook, 2006) resolves dyadic dependencies by estimating individual pathways simultaneously:*

- Actor Effects (a): *The influence of your own predictor status on your own target outcome (e.g. Own connection → Own satisfaction).*
- Partner Effects (p): *The interpersonal crossover of your partner's predictor on your outcome (e.g. Partner's connection → Own satisfaction).*
- Simultaneous Control: *Modeling actor and partner paths together extracts true cross-partner influence while isolating individual biases.*



# Indistinguishable vs. Distinguishable

## Indistinguishable Dyads

*Members are exchangeable. No baseline or stable structural variable differentiates them.*

- Arbitrary Labels: *Partner labels (e.g. A vs B, Left vs Right) are completely interchangeable.*
- Equality Constraints: *Actor and partner slopes must be constrained to be mathematically equal across partners.*
- Typical Examples: *Same-sex roommates, co-workers of equal organizational rank, sibling pairs.*

## Distinguishable Dyads

*Members can be classified using a systematic, theoretical variable that is consistent across the sample.*

- Systematic Labels: *Partners are labeled using a meaningful criterion (e.g. Wife/Husband, Boss/Employee).*
- 
- Typical Examples: *Heterosexual couples, medical caregiver-patient dyads, manager-employee pairs.*

Important: *Indistinguishability is an empirical hypothesis. Always test it formally (by comparing a constrained model with an unconstrained version) before locking in equal-parameter constraints!*

# Modeling Options Taxonomy

*Choose the optimal modeling combination based on your data layout, distinguishability, and nested structures:*

| *Modeling Strategy* | *Analytical Framework* | *Data Layout* | *Core Application & Mechanism* |
|---|---|---|---|
| Indistinguishable APIM | *MLM (e.g., lme4)* | *Long Format* | *Symmetric pooling of slopes across members; simple intercept ICC check.* |
| Moderated MLM APIM | *MLM (e.g., lme4)* | *Long Format* | *Incorporates member interactions directly into continuous pathways.* |
| Two-Intercept MLM APIM | *MLM (e.g., lme4)* | *Long Format* | *Suppresses overall intercept to map distinct group means simultaneously.* |
| Standard SEM APIM | *SEM (e.g., lavaan)* | *Wide Format* | *Strict parameter constraints, goodness-of-fit indices, FIML for missing data.* |
| Multilevel SEM (MSEM) | *MSEM (e.g., lavaan)* | *Long (Nested)* | *Simultaneous latent variables and complex nesting within nested arrays.* |

Framework Choice: *MLM models excel at complex nesting patterns, daily diaries, and unbalanced structures. SEM provides fit indices, simultaneous constraints, and mediation tests.*

Specification Tip: *Wide-format SEM allows testing of complex, non-linear partner spillover parameters and structural residuals that are difficult to write in traditional MLM notation.*

# Reshaping Dyadic Data

## ↔ Wide Format (1 Row/Dyad)

*Standard for SEM with suffix variable labels mapping to specific roles.*

| dyad_id | wnc_a | wnc_p | sat_a | sat_p |
|---|---|---|---|---|
| 101 | 4.2 | 3.8 | 5.1 | 4.9 |
| 102 | 3.1 | 3.2 | 4.8 | 4.7 |

```
# Reshape from Wide to Long
long_df <- wide_df %>%
  pivot_longer(cols = -dyad_id,
    names_to = c(".value", "role"),
    names_sep = "_")
```

## ↕ Long Format (2 Rows/Dyad)

*Standard for MLM with repeated observations nested within a single key.*

| dyad_id | role | wnc | sat |
|---|---|---|---|
| 101 | person_1 | 4.2 | 5.1 |
| 101 | person_2 | 3.8 | 4.9 |
| 102 | person_1 | 3.1 | 4.8 |
| 102 | person_2 | 3.2 | 4.7 |

```
# Reshape from Long to Wide
wide_df <- long_df %>%
  pivot_wider(id_cols = dyad_id,
    names_from = role,
    values_from = c(wnc, sat))
```

# Multilevel Modeling (Long Format)

*To handle indistinguishable datasets, format the file in long format (one row per person, two rows per dyad). Random effects adjust for nested interdependence.*

- Empty Intercept-Only Model: *Compute the Intra-class Correlation (ICC) to estimate the proportion of shared variance.*
- Model Specification: *Estimate both actor and partner paths symmetrically, pooling slope estimators across the dyad members.*
- Centering Strategy: *Group-mean center independent variables to cleanly separate within-dyad from between-dyad effects.*

## MLM Formulation

```
satisfaction_ij = (β_0 + u_j) + β_1(wnc_ij) + β_2(partner_wnc_ij) + e_ij
```

*Where i represents the individual, j represents the dyad, u_j represents the dyad-level random intercept, and e_ij is individual residual.*

*Requires the lme4 package in R for model fitting, and lmerTest to compute Satterthwaite degrees of freedom.*

# R Code: Indistinguishable MLM

*We estimate the basic APIM framework step-by-step using lmer. Ensure your package environment is properly configured:*

- *Loads data in* long format *(one row per partner).*
- *Calculates the **ICC** to verify within-dyad correlation.*
- *Includes covariates: Recovery, Children status, and Dual-Earner category.*

> Note: *In indistinguishable setups, slope estimates are pooled across groups.*

```
# Indistinguishable MLM (script 02)
library(lme4); library(lmerTest)
load("dyad_data.RData")

# 1. Null model + ICC
m0 <- lmer(sat ~ 1 + (1 | dyad), data=ddl)
icc(m0)  # between / (between + residual)

# 2. Full APIM
m1 <- lmer(sat ~ wnc + partner_wnc +
          recovery + has_children +
          dual_earner + (1 | dyad), data=ddl)
summary(m1)
```

# Hands-On: Indistinguishable MLM

## Active Code Session

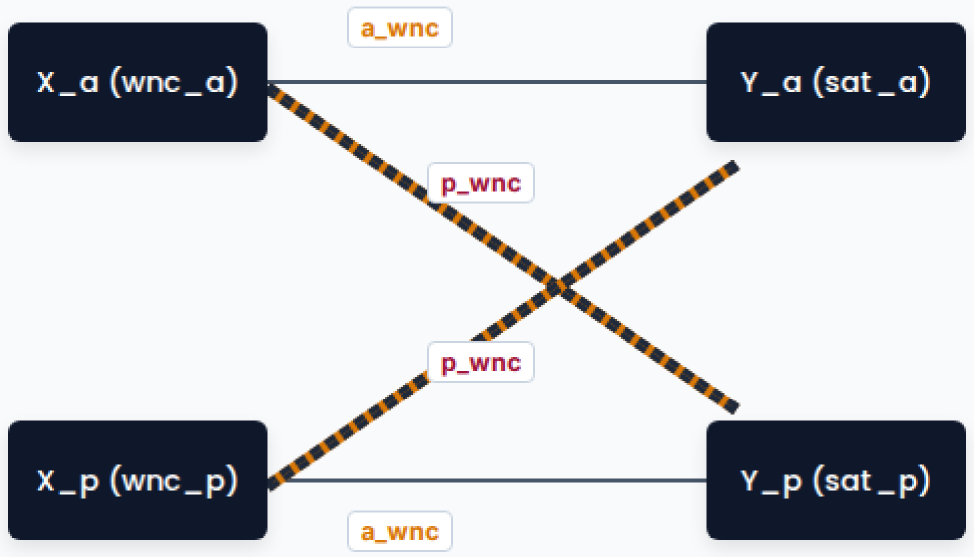
*Let's execute your first multilevel APIM scripts in RStudio:*

- Open: *scripts/02_indistinguishable_mlm.R*
- Run: *Estimate the null model and compute the ICC:*
  *ICC = $\sigma^2$_between / ($\sigma^2$_between + $\sigma^2$_residual)*
- *Run the full APIM*

# Structural Equation Modeling

*We specify explicit path lines inside wide-format data (one row per dyad with distinct suffix labels _a / _p). Equality constraints are declared to enforce indistinguishability:*

- Slope Constraints: *Own paths (b_a) and partner paths (b_p) are labeled identically.*
- Interceptions/Variances: *Model intercepts, variances, and residuals are restricted to equality.*



# R Code: Indistinguishable SEM

*We write structural code using lavaan syntax. Suffix tags label identical parameters across equations to enforce symmetry:*

- **Label Syntax:** *Prefixing coefficients with label names (e.g. a_wnc*wnc_a) forces equality.*
- **Interceptions & Variances:** *Restricting intercepts (alpha*1) and residuals (var_res*sat_a) establishes full indistinguishability.*

```
# Indistinguishable SEM wide (script 03)
library(lavaan)
mod_c <- '
  sat_a ~ a_wnc*wnc_a + p_wnc*wnc_p +
         a_rec*rec_a + p_rec*rec_p +
         c_ch*child + c_du*dual_earner
  sat_p ~ a_wnc*wnc_p + p_wnc*wnc_a +
         a_rec*rec_p + p_rec*rec_a +
         c_ch*child + c_du*dual_earner
  sat_a ~ alpha*1
  sat_p ~ alpha*1       # equal intercepts
  sat_a ~~ var_res*sat_a
  sat_p ~~ var_res*sat_p  # equal residuals
'
fit <- sem(mod_c, data=ddw)
```

# Hands-On: Indistinguishable SEM

## </> Active Coding Session

*Let's transition from multilevel modeling to structural equation modeling:*

- Open: *scripts/03_indistinguishable_sem.R*
- Evaluate: *Review the three alternative analytical paradigms defined in the script:*
  - *Model A: Cluster-robust Standard Errors (quick, long format)*
  - *Model B: Two-level SEM (within/between decomposition)*
  - *Model C: Wide-format APIM (primary, with full equality constraints)*

## Questions to Explore

*Compare the parameter values from Model A (cluster-robust) with Model B (two-level SEM) to Model C (wide-format constraint).*

*How does structural estimation impact statistical precision in small-to-medium samples?*

# Core Analytical Insights

### Method Convergence

*Balanced data profiles yield identical point estimates in both **Multilevel (long)** and **SEM (wide with equal constraints)** designs.*

### LRT Testing

*Perform a Chi-Square Difference Test comparing the fully constrained model with an unconstrained version to empirically test for indistinguishability.*

### Analytical Trade-offs

*Why choose SEM over Multilevel for indistinguishable dyads?*

- ✓ ***Supports multi-item latent variables.***
- ✓ ***FIML retains dyads with partial missing values.***
- ***Supports full mediation models.***

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# Moderator Interaction Modeling

*When dyad members are distinguishable (e.g. heterosexual couples), we can model interactions between predictors and the grouping factor (e.g. gender) in long format.*

### Version A: Simple Moderator (Conflated)

```
satisfaction ~ wnc * gender + recovery * gender + (1 | dyad_id)
```

### Version B: Full APIM Moderator (Decomposed)

```
satisfaction ~ wnc * gender + partner_wnc * gender + recovery *
gender + partner_recovery * gender + (1 | dyad_id)
```

### ⚠ Warning for crossover effects

*Simple moderator setups (Version A) conflate actor and partner pathways. This can lead to misleading conclusions when evaluating crossover effects.*
*Always specify the **Full APIM Moderator** (Version B) to properly separate actor from partner pathways when using the moderator approach in long format.*

# R Code: Moderator MLM

*We write the interaction script for Version B using lmer. This model evaluates if actor and partner effects differ significantly by gender:*

- Reference Group: *Male is set as the reference group (simple slope = b_wnc).*
- Interaction Term: *Estimates the difference in slopes (female simple slope = b_wnc + b_wnc:genderfemale).*

*See script 04*

```
# Distinguishable: MLM Moderator (script 04)
library(lme4); library(lmerTest)
load("dyad_data.RData")

# Baseline (no interactions)
m_base <- lmer(sat ~ wnc + partner_wnc +
               rec + p_rec + child + dual +
               (1 | dyad), data=ddl)

# Full APIM with gender interactions
m_mod  <- lmer(sat ~ wnc*gender +
               partner_wnc*gender + rec +
               p_rec + child + dual +
               (1 | dyad), data=ddl)
summary(m_mod)
anova(m_base, m_mod)
```

# Hands-On: Moderator APIM

## Active Coding Session

*Let's practice modeling interaction terms in distinguishable dyads:*

- **Open:** *scripts/04_distinguishable_mlm.R*
- **Analyze:** *Fit the full APIM moderator interaction model in R*
- **Evaluate:** *Identify the interaction terms: wnc:genderfemale and partner_wnc:genderfemale. Calculate the simple slopes for males and females.*

## Theoretical RQ Example

*"Does gender moderate the crossover of work-family connection between spouses, such that the crossover effect of partner connection on relationship satisfaction is stronger for females than for males?"*

***Are the simple slopes statistically significant for both males and females?***

# The Two-Intercept Model

*To avoid setting an arbitrary reference group (like males), we can suppress the global intercept using the 0 + gender syntax.*

- Direct Parameterization: *Directly estimates intercepts and slopes for both males and females, simplifying group comparisons.*
- Flexible Equality Constraints: *Easily constrain slopes to be equal across groups while still estimating separate intercepts.*

## Three Modeling Stages

*Compare these nested models using Likelihood Ratio Tests:*

- Model 1: *Equal intercepts, equal slopes (most constrained model).*
- Model 2: *Different intercepts, equal slopes (recommended baseline model).*
- Model 3: *Different intercepts, gender-varying slopes (full interaction model).*

# R Code: Two-Intercept MLM

*The two-intercept model is specified in lme4 by suppressing the global intercept (using 0 + gender) so each group gets its own intercept:*

- **Interpretable Outputs:** *Coefficients are presented as separate intercepts (gendermale, genderfemale) rather than standard difference offsets.*
- **Nested Model Comparisons:** *Compares nested model equations to identify distinct parameter slopes.*

*See script 05*

```
# Two-Intercept MLM (script 05)
library(lme4); library(lmerTest)
load("dyad_data.RData")

# Equal slopes, separate intercepts
m2 <- lmer(sat ~ 0 + gender + wnc +
           partner_wnc + rec + p_rec +
           child + dual_earner +
           (1 | dyad), data=ddl)
summary(m2)

# Coefficients: gendermale, genderfemale
# are the two dyad-member intercepts
```

# Distinguishable: SEM Moderator

## The Approach

*Mirror the lme4 gender moderator (script 06) as a two-level lavaan model. Gender interactions enter at the within-dyad level; dyad-level covariates (children, dual earner) at the between level.*

## lavaan Model

***Two-level SEM: gender interactions are placed at the within-dyad level. Numeric coding (1=male, 2=female) makes the simple-slope labels explicit:***

```
ss_wnc_M  := a_w + aw_int * 1
ss_wnc_F  := a_w + aw_int * 2
```

**Tip:** ***Models are saturated at the within level, so chi-square difference tests are not available. Use simple-slope labels (:=) to interpret the moderation.***

# R Code: SEM Moderator

### </> Two-Level SEM with Gender Moderation

*We construct a two-level lavaan model that mirrors the lmer interaction (script 06):*

```
# SEM Moderator (script 06)
library(lavaan)
ddl$gender_num <- as.numeric(ddl$gender)
ddl$wnc_x_g    <- ddl$wnc * ddl$gender_num
ddl$pw_x_g     <- ddl$partner_wnc * ddl$gender_num

mod <- '
  level: within
    sat ~ a_w*wnc + p_w*partner_wnc +
         ar_w*recovery + pr_w*partner_recovery +
         g_w*gender_num +
         aw_int*wnc_x_g + pw_int*pw_x_g
  level: between
    sat ~ c_ch*has_children + c_du*dual_earner
  # Simple slopes
  ss_wnc_M  := a_w + aw_int * 1
  ss_wnc_F  := a_w + aw_int * 2
'
fit <- sem(mod, data=ddl, cluster="dyad_id")
summary(fit)
```

# Distinguishable: SEM (lavaan)

*In structural equation modeling, distinguishable dyads are modeled using wide-format datasets. We can estimate separate slopes, variances, and covariances for each group without constraints:*

- Unconstrained Baseline: *Establishes unique pathways (e.g. males: a_m, females: a_f).*
- Nested Constraints: *Constrain slopes to be equal (e.g. a == a) to test for gender differences via a Chi-Square Difference Test.*

```
# Distinguishable SEM wide (script 07)
library(lavaan)
# Three nested models:
#  1) Unconstrained  - all paths free
#  2) Slopes equal   - labels a,p,...
#  3) Fully equal    - + equal intercepts
mod_un <- '
  sat_a ~ a_h*wnc_a + p_h*wnc_p + ...
  sat_p ~ a_w*wnc_p + p_w*wnc_a + ...
  sat_a ~ int_a*1
  sat_p ~ int_p*1   # separate intercepts
'
fit_un <- sem(mod_un, data=ddw)
lavTestLRT(fit_un, fit_slopes,
           fit_full)  # nested LRTs
```

# Hands-On: Distinguishable SEM

## Active Coding Session

*Let's practice setting constraints in structural equation models:*

- Open: *scripts/06_distinguishable_sem_wide.R*
- Execute: *Fit the unconstrained, slopes-equal, and fully-equal (slopes + intercepts) models.*
- Compare: *Run Chi-Square Difference Tests comparing the three models to determine if constraining male and female slopes to be equal significantly worsens model fit.*

## Core Modeling Decisions

*"If the Chi-Square Difference Test is non-significant, adopt the constrained model for parsimony. If significant, retain the unconstrained model to allow for gender differences."*

***Does model fit change significantly?***

# Distinguishable: Model Comparison

*Compare the three primary structural modeling methods used for distinguishable dyad analyses:*

| *Feature* | *Moderator Approach (lme4)* | *Two-Intercept (lme4)* | *SEM Approach (lavaan)* |
|---|---|---|---|
| Data Format | *Long Format (One row/person)* | *Long Format (One row/person)* | *Wide Format (One row/dyad)* |
| Intercepts | *Reference Group + Offset* | *Two Direct Intercepts* | *Two Direct Intercepts* |
| Testing Slope Differences | *Via Interaction Significance* | *Via Likelihood Ratio Test* | *Via Chi-Square Difference* |
| Fit Indices | *No (relies on log-likelihood)* | *No (relies on log-likelihood)* | *Yes (CFI, RMSEA, AIC/BIC)* |
| Missing Data | *Listwise Deletion* | *Listwise Deletion* | *FIML (retains partial dyads)* |
| Best Application | *Testing external continuous moderators* | *Direct means comparisons* | *Complex mediation & latent vars* |

# The Buffer Hypothesis

*We replicate the theoretical model of **Hahn, Binnewies, & Dormann (2014)** to examine if dyad-level moderators alter standard APIM pathways:*

- Research Question: *Does the presence of children moderate the partner crossover effect (Partner Connection → Own Satisfaction)?*
- Theoretical Mechanism: *Partner crossover represents stress transmission. Joint domestic demands (e.g. child-rearing) may alter these crossover pathways.*
- Findings: *The crossover effect is buffered (weaker) for couples with children, as shared family responsibilities alter cognitive work-life boundaries.*

## Moderated APIM Structure

*The model includes interaction terms between the **Partner Connection** predictor and the **Has Children** indicator.*

*Both MLM and SEM frameworks can accommodate dyad-level moderators.*

# Multilevel vs. SEM Formulations

*We estimate the interaction using either lme4 or lavaan:*

- **lme4 Interaction:**
  *wnc * has_children + partner_wnc * has_children*
- **lavaan Interaction:** *Requires creating manual product indicators for the interactions in wide format before fitting.*

</> **Open:** *scripts/08_moderated_apim.R. Run the script and plot the interaction slopes.*

```r
# Moderated APIM, MLM (script 08)
library(lme4); library(lmerTest); library(interactions)

m_mod <- lmer(satisfaction ~ wnc + partner_wnc +
                has_children +
                wnc:has_children + partner_wnc:has_children +
                (1 | dyad_id), data = ddl)

interact_plot(m_mod, pred = partner_wnc, modx = has_children)

# *** SEM: lavaan ***

# SEM Approach: lavaan (script 08)
ddw$wnc_a_children <- ddw$wnc_a * ddw$has_children
ddw$wnc_p_children <- ddw$wnc_p * ddw$has_children

mod <- '
  sat_a ~ a*wnc_a + p*wnc_p + a_int*wnc_a_children
  sat_p ~ a*wnc_p + p*wnc_a + p_int*wnc_p_children
  actor_no_child  := a + a_int * 0
  actor_has_child := a + a_int * 1
'

fit <- sem(mod, data = ddw)
summary(fit)
```

# Advanced: APIM Power Analysis

*Dyadic power analysis is complex because power is a function of the **Intra-class Correlation (ICC)**, effect size, and the number of dyads (not individuals).*

- Basic APIM: *Requires a minimum of 50 dyads; N = 100-120 dyads is preferred.*
- Partner Effects: *Require larger samples than actor effects due to shared variance.*
- Moderation: *Cross-level interactions typically require N > 150 dyads.*

## Sample Requirements Metric

*$\alpha$ = 0.05, power = 0.80 — required N (dyads):*

| Effect size (β) | β = .15 | β = .25 |
|---|---|---|
| Indistinguishable *dyads* | 159 | 59 |
| Distinguishable *dyads* | 348 | 129 |

*Estimated with APIMPowerR:*
*https://robert-a-ackerman.shinyapps.io/APIMPowerRdis/*

# Three-Level Nesting Structures

*Dyads are often nested within larger clusters (e.g. organizations, departments, clinic teams), or measured over time:*

- 3-Level syntax in lme4:
  *satisfaction ~ wnc + (1 | team_id) + (1 | dyad_id)*
- Daily Diary / ESM Designs: *Days nested within Persons, nested within Dyads.*
  - *Allows modeling of daily actor/partner lag fluctuations.*
  - *Time-lagged effects: Does yesterday's connection predict today's relationship satisfaction?*

## Bayesian Estimations

*For small samples or complex models with multiple random slopes, Bayesian estimations provide stable parameter estimates:*

- brms: *Stan-based package with intuitive, lme4-like syntax.*
- blavaan: *Fits Bayesian SEM models.*

# Missing Data in Dyadic Designs

*If one partner has missing data, listwise deletion in standard regression removes the entire dyad. This significantly reduces statistical power:*

- **SEM FIML:** *Full Information Maximum Likelihood estimates parameters using all available data under MAR assumptions. This is the preferred approach.*
- **Multiple Imputation:** *Impute missing values at the dyad level using packages like mice to preserve dyadic dependencies.*

# Questions & Discussion

## Thank You!

***We have successfully walked through the foundations, estimations, and advanced extensions of the Actor-Partner Interdependence Model (APIM) in R.***

🗁 **R Workspace Datasets & Scripts:** ***Check the local scripts/ directory to run exercises.***